

turboswarm: A general-purpose, extensible Particle Swarm Optimization library with a Rust core

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June 2026

Abstract

turboswarm is a general-purpose, open-source Particle Swarm Optimization (PSO) library with a compute core written in Rust and an ergonomic Python API. It targets real, integer, mixed, binary and interval-valued (“grey”) decision variables, and exposes established algorithmic variants, swarm topologies, constraint handling, multi-objective optimization, and visualization behind a single, uniform interface. The optimization loop is fully decoupled from any concrete variant through three traits (**SearchSpace**, **Velocity**, **Topology**), so adding a new variant is a self-contained trait implementation rather than a fork of the core. The same Rust core is usable directly from Rust through zero-cost generics and from Python through PyO3 and **maturin**, where native objectives run without holding the Global Interpreter Lock. Every experiment is seedable and therefore reproducible. On a five-function benchmark across three dimensions, the compiled core is the fastest among the libraries compared, with a measured speedup of roughly 2–3.5× over the fastest pure-Python competitor at comparable solution quality. The library is distributed on PyPI (**turboswarm**) and crates.io (**turboswarm-core**), documented, and archived on Zenodo (DOI 10.5281/zenodo.20832446).

1 Introduction

Particle Swarm Optimization [5] is a population-based, gradient-free optimization method in which a swarm of candidate solutions (“particles”) moves through the search space guided by each particle’s own best position and the best position found in its neighbourhood. It is one of the most widely used metaheuristics for continuous and combinatorial optimization, with applications across engineering, machine learning and the sciences.

Researchers and students need a PSO implementation that is fast, reproducible, broad in scope, and easy to extend with new variants for comparative studies. Existing Python libraries cover parts of this need, but they are implemented predominantly in pure Python, which ties the cost of the optimization loop to the interpreter and makes large or repeated experiments slow. **turboswarm** was written to provide a compiled, reproducible and extensible PSO toolkit that is nonetheless as easy to call as a pure-Python package, suitable for research experiments, algorithm benchmarking and teaching.

2 State of the field

Several mature open-source PSO and metaheuristics packages exist, including **pyswarms** [7], **pymoo** [1], **DEAP** [4] and **nevergrad** [8]. These are valuable and widely adopted, but they are implemented

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mainly in pure Python, so the swarm loop itself runs in the interpreter. `turboswarm` differs in three ways that motivated building a new library rather than contributing to an existing one. First, its optimization loop and native objectives are compiled (Rust) and run without the Global Interpreter Lock (GIL), giving a measured speed advantage (Section 6). Second, it treats extensibility as a first-class design goal: a new variant is a self-contained trait implementation, not a fork of the core (Section 3). Third, it provides a grey-number (interval-valued) search space, in which each decision variable is known only to lie within an interval — a capability we are not aware of in the libraries above. Together these make `turboswarm` complementary to, rather than a re-implementation of, the existing ecosystem.

3 Software design

The optimization loop is fully decoupled from any concrete variant through three traits.

- **SearchSpace** defines the domain. The only difference between real and integer variables lives in its `decode` step, so the swarm always operates on a continuous internal representation ($\mathbb{R}^n$) and discretization happens only at evaluation time.
- **Velocity** defines the update rule, so a single PSO variant — inertia [9], constriction [2] or the Fully Informed Particle Swarm [6] — is exactly one implementation of this trait.
- **Topology** defines the social structure of the swarm: global (gbest), ring (lbest), Von Neumann or random.

Because the loop is generic over these traits, adding a variant requires no change to the core and is accompanied by a convergence test against a known optimum. For the Python boundary the same traits are implemented for boxed trait objects, so variants can be selected at runtime by name without duplicating the loop. This design confines the real/integer/mixed/binary/grey distinction to the search space and makes the library a convenient base for comparing variants under identical conditions.

4 Features

- **Velocity variants:** inertia [9], constriction [2] and Fully Informed PSO [6].
- **Topologies:** global (gbest), ring (lbest), Von Neumann and random.
- **Search spaces:** continuous, integer (with a discretization strategy), binary, mixed (per-dimension type) and grey (interval-valued variables, optimized over their centre and spread).
- **Multi-objective (MOPSO):** an external Pareto archive with a turbulence operator, crowding-distance or Coello’s adaptive grid [3] for diversity, and a hypervolume quality indicator.
- **Constraints:** inequality and equality constraints via a penalty, plus an optional repair operator that maps candidates back to the feasible set.
- **Run control:** stopping on a target value, evaluation budget, wall-clock budget or stagnation, with a per-iteration callback and a reported stop reason.
- **Boundary handling:** clamp, reflect, wrap or reinitialize.

- **Performance:** velocity clamping, `rayon`-based parallel evaluation and a NumPy-vectorized batch path.
- **Hyperparameter sensitivity:** a Cartesian-product sweep over hyperparameter values, aggregated over seeds.
- **Benchmarks:** sphere, rastrigin, rosenbrock, ackley, griewank, schwefel and the CEC-family functions (bent cigar, discus, elliptic, zakharov, levy, expanded Schaffer), each with metadata (recommended bound, known optimum).
- **Visualization (Python):** convergence curves, variant comparison, 2D and 3D swarm animation, Pareto-front and sensitivity plots.
- **Integrations:** a SciPy `minimize` drop-in, a scikit-learn `PSOSearchCV`, an Optuna sampler, pandas export and parallel-objective backends.

5 Usage

From Python, an objective can be a callable or the name of a native benchmark; bounds are given per dimension or as a single pair broadcast over `dim`.

```
import turboswarm as pso

# Native benchmark, runs in Rust without the GIL
r = pso.minimize("rastrigin", bounds=(-5.12, 5.12), dim=2, seed=42)

# A user objective, a chosen variant and topology
r = pso.minimize(lambda x: sum(xi*xi for xi in x),
                 bounds=[(-5, 5)] * 3, velocity="fips", topology="ring", seed=1)
print(r.best_position, r.best_value)
```

The same core is callable from Rust through zero-cost generics:

```
use turboswarm_core::prelude::*;
use turboswarm_core::benchmarks::rastrigin;

let space = ContinuousSpace::uniform(2, -5.12, 5.12);
let velocity = InertiaVelocity::new(0.9, 1.49445, 1.49445).with_decay(0.4);
let params = PsoParams { seed: Some(42), ..Default::default() };
let result = Pso::new(space, velocity, GlobalBest::new(), params).minimize(rastrigin);
```

6 Performance

To illustrate the benefit of the compiled core we compare `turboswarm` against `pyswarms` [7], `pyswarm`, `pymoo` [1] and `DEAP` [4] on five standard functions (sphere, rastrigin, ackley, rosenbrock, griewank) and three dimensions (2, 10, 30), using an identical swarm configuration for every library (40 particles, 200 iterations, $w = 0.729$, $c_1 = c_2 = 1.49445$; median over five seeds after a warm-up). Through its native objective path `turboswarm` is the fastest library across all functions and dimensions tested. Figure 1 shows its speedup over `pyswarms`, the fastest competitor: roughly $3.5\times$ at two dimensions, narrowing towards $2\times$ at thirty dimensions as the vectorized NumPy objective amortizes more of the per-iteration overhead. Solution quality was comparable across libraries; `turboswarm` reached the best objective in several cases, while `pymoo` did so on a

few high-dimensional multimodal problems. DEAP, a general-purpose evolutionary-computation toolkit that provides a PSO recipe rather than a packaged solver (which we adapted to the same inertia rule and configuration), is the slowest baseline because its interpreter-bound, non-vectorized loop carries the most per-iteration overhead: **turboswarm**’s native path is a median $12\times$ faster (roughly $7\text{--}15\times$ across functions and dimensions) at comparable solution quality.

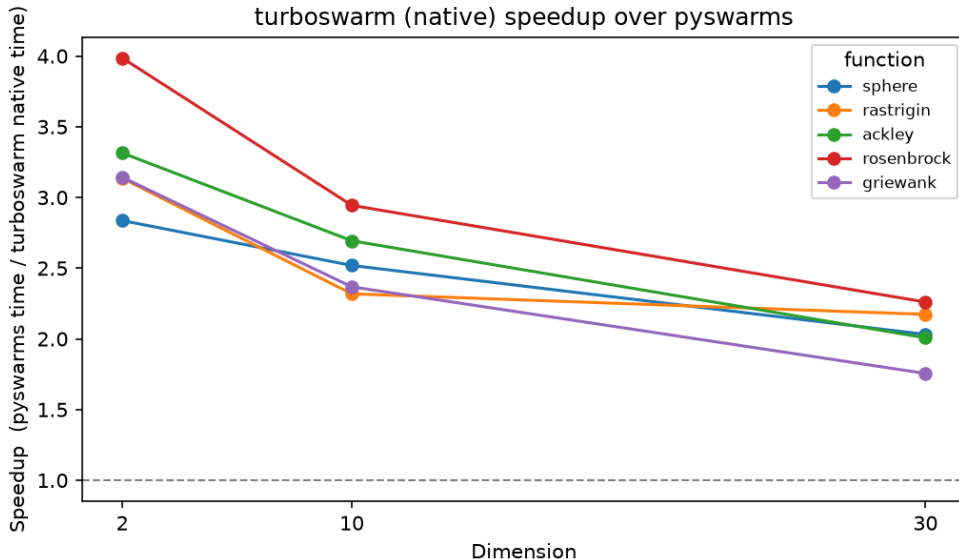


Figure 1: Speedup of **turboswarm** (native objective path) over **pyswarms** across dimensions, for five benchmark functions.

Measurements were taken on an Apple M5 (10 cores, macOS 26.5). Absolute numbers are machine-dependent; the full benchmark — including the raw results and machine provenance — is reproducible with `benches/bench_suite.py`.

7 Availability and citation

turboswarm is released under the MIT license. The Python package is on PyPI as **turboswarm** (`pip install turboswarm`) and the Rust core on crates.io as **turboswarm-core**. The source code, documentation and issue tracker are at <https://github.com/turboswarm/turboswarm.github.io>, with a documentation portal at <https://turboswarm.github.io>. Each release is archived on Zenodo under the concept DOI 10.5281/zenodo.20832446. If you use **turboswarm** in academic work, please cite this paper and the archived software release.

8 Conclusion

turboswarm provides a compiled, reproducible and extensible PSO toolkit that is as easy to call as a pure-Python package. By exposing many PSO variants, topologies and search spaces — including an interval-valued “grey” space — behind one reproducible interface, it lowers the cost of comparative algorithmic studies and supports teaching through its built-in visualization. Its trait-based core is also the foundation for ongoing methodological work on optimization with grey (interval-valued) variables.

AI usage disclosure

The `turboswarm` library itself — the Rust core, the Python bindings and the algorithms — was designed and written by the author. Generative AI (Anthropic’s Claude, used through Claude Code) assisted with drafting this paper, implementing the comparative benchmark suite, and editing parts of the documentation. All AI-assisted text and code were reviewed, tested and validated by the author, who takes full responsibility for the content.

Acknowledgements

The author received no specific funding for this work, and acknowledges the open-source Rust and PyO3 communities, whose tooling underpins the Rust–Python boundary used here.

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